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Stratigraphy, lithology, paleomagnetism, and fission track ages of some ignimbrite formations in the Matahana Basin, New Zealand

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Abstract Three new ignimbrite formations are described from the Matahana Basin, southwest of Rotorua, and are formally named the **Pukerimu Ignimbrite, Tikorangi Ignimbrite,** and **Rahopaka Ignimbrite**. All are older than 0.72 Ma. This is confirmed by fission track dating of zircons extracted from 2 of these ignimbrites and additional dates on younger horizons. The Tikorangi Ignimbrite is dated at 1.03 ± 0.18 Ma, Rahopaka Ignimbrite at 0.72 ± 0.16 Ma, Marshall Ignimbrite at 0.52 ± 0.14 Ma, and the Mamaku Ignimbrite at 0.14 ± 0.08 Ma.

Keywords Pukerimu Ignimbrite; Tikorangi Ignimbrite; Rahopaki Ignimbrite; Marshall Ignimbrite; new stratigraphic names; Matahana Basin; paleomagnetism; fission track dating

INTRODUCTION

The Matahana Basin lies 19 km southwest of Rotorua on the western margin of the Taupo Volcanic Zone and extends over an area of 70 km^2 between the Mamaku Plateau in the northwest, and the Guthrie Graben in the southeast. Grindley (1959) mapped the geology of Sheet N85 (1:63 360), which includes the Matahana Basin, and grouped all the ignimbrites as either Ohakuri Formation or Waiotapu Ignimbrite. The whole area is shown in Healy et al. (1964). The stratigraphy of the younger ignimbrites was established by Martin (1961) and Blank (1965), but no further detailed work has been done in the area.

Detailed mapping was carried out by R. P. M. and full descriptions of all the ignimbrites are given in Murphy (1977). In this paper, 3 previously unknown mappable ignimbrites and 1 redefined ignimbrite are described, and new age data are presented.

STRATIGRAPHY

Most of the rocks exposed in the area, dominantly rhyolitic ignimbrite, tephra, and lava domes, are of primary volcanic origin. Lacustrine beds, alluvial scree, and valley deposits form only a minor part of the sequence.

The major ignimbrites are flat lying; the 3 previously undescribed ignimbrites form the base of the sequence (Fig. 1). They are well-defined mappable units which appear not to be correlated with ignimbrite formations described elsewhere.

PUKERIMU IGNIMBRITE (pi) (new name)

This is the oldest exposed ignimbrite in the Matahana area. No type section has been assigned as exposure is poor and the base of the ignimbrite was not seen, but a reference section is located 0.6 km from the Pukerimu–Tikorangi Roads junction between N85/545878* and 555885. Here the ignimbrite is flat lying, 79+ m thick, and contains plagioclase feldspar as the dominant phenocryst with very minor hypersthene and no quartz. The degree of welding decreases upwards such that the top is only partially welded. There is also an upward decrease in the abundance of rhyolite xenoliths, and pumice clasts change from grey white to orange. The upper part of the Pukerimu Ignimbrite is also exposed within the Matahana Basin at N85/570885; here it is strongly welded and orange-brown. The Pukerimu Ignimbrite has a gradational upper contact with the overlying Tikorangi Ignimbrite, and together they form a compound cooling unit.

TIKORANGI IGNIMBRITE (ti) (new name)

The type section is located along Tikorangi Road, 300 m north of the Pukerimu–Tikorangi Roads junction between N85/547876 and 545878. The

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*Grid references are based on the national thousand-yard grid of the 1:63 360 topographical map series (NZMS 1).

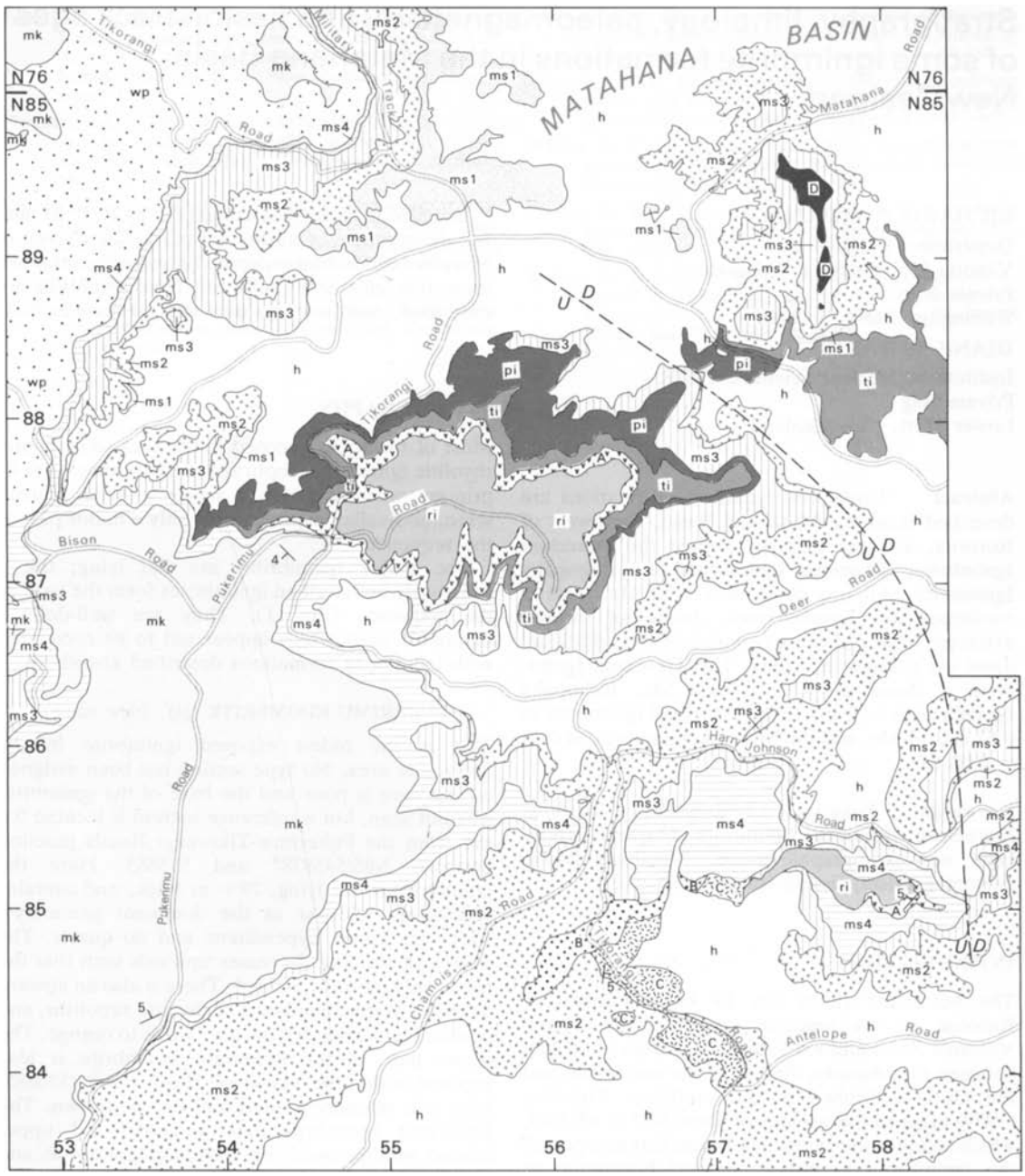
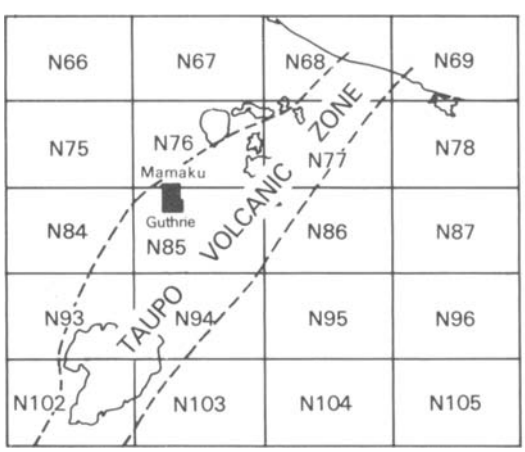
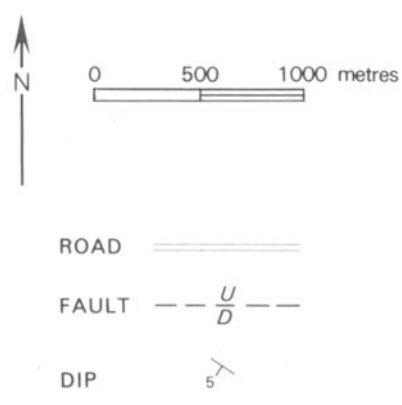


Fig. 1 Geological map of the study area. Base map taken from NZMS 1, 1 : 63 360 topographical map sheets N76—Rotorua and N85—Waiotapu.

GEOLOGICAL LEGEND FOR FIG. 1

SYMBOL	FORMATION	LITHOLOGY	AGE (Ma)
h	Undifferentiated volcanics and alluvium	Bedded fluviatile pumice, rhyolite, and quartz sands and boulders, slope wash	
mk	Mamaku Ignimbrite	Grey–brown quartzose ignimbrite	0·14
D	Unit D	Quartzose pumiceous breccia	
wp	Waiotapu Ignimbrite	Densely welded quartz–poor lenticulite	
ms4	Marshall Ignimbrite unwelded	Poorly welded, crystal–poor, pumice breccia	0·52
ms3	welded		
ms2	unwelded		
ms1	pumice fall		
C	Unit C	Interbedded breccia and hard stratified lake silts	
B	Undifferentiated pyroclastics B	Densely welded crystal–rich ignimbrites and pisolitic tuff	
ri	Rahopaka Ignimbrite	Grey–brown moderately welded hornblende–rich ignimbrite	0·72
A	Undifferentiated pyroclastics A	Ignimbrite, tuff, breccia and lacustrine silts	
ti	Tikorangi Ignimbrite	Grey–black densely welded quartz lenticulite	1·03
pi	Pukerimu Ignimbrite	Pumiceous crystal–poor, quartz–free ignimbrite	





Martin 1961		This study	
Marshall Ignimbrites	Pumice–tuff breccia	Marshall Ignimbrite	unwelded
	Taupaki Ignimbrite		welded
	Tuff–breccia ignimbrite		unwelded
			Pumice–fall
	Lenticulitic ignimbrite	?	

Fig. 2 Nomenclature of the Marshall Ignimbrite sequence.

ignimbrite is flat lying, 30 m thick, and the contact with the underlying Pukerimu Ignimbrite is gradational. A dark, densely welded lenticulite containing large black fiamme is at the base, grading upwards to a grey, densely welded ignimbrite containing randomly oriented black pumice clasts (<150 mm) at the top. The basal lenticulite is best exposed in a reference section within the Matahana Basin, 2.3 km northeast of the type section (N85/573884).

The ignimbrite is generally deficient in crystals; phenocrysts include plagioclase, magnetite, hypersthene, calcium-poor clinopyroxene, and hornblende. Quartz is confined to the upper third of the flow. Small rhyolite xenoliths are common.

The full extent of the ignimbrite is unknown as much was removed by erosion prior to eruption of the Marshall Ignimbrite, and present outcrop is restricted by the cover of younger pyroclastics and alluvium. However, it does extend south from the Matahana Basin towards the Guthrie Graben where it is presumably downthrown by northwest-trending faults. On Harry Johnson Road (N85/584854) a basalt scoria deposit contains xenoliths of Tikorangi Ignimbrite indicating that it is present at depth in this area.

A series of pyroclastic deposits overlies the Tikorangi Ignimbrite, and these are described elsewhere (Murphy 1977).

RAHOPAKA IGNIMBRITE (ri) (new name)

The type section of the Rahopaka Ignimbrite is a 300 m roadside section extending east from the Pukerimu–Tikorangi Roads junction at N85/546873 to 550874. The ignimbrite is flat lying, 60+ m thick, and overlies pyroclastic deposits. At the type section the lower contact of the Rahopaka Ignimbrite is obscured by a slip, but above this it is grey-brown, crystal rich, and densely welded near the base. The degree of welding decreases towards the top where it is generally only moderately welded. Phenocrysts in decreasing order of abundance are plagioclase, quartz, hornblende, magnetite, and hypersthene. The visible abundance of green-brown hornblende is the most characteristic mineralogical feature of this ignimbrite. The ignimbrite base is also exposed on Rusa Road, 1.8 km from the Tikorangi–Rusa Roads junction (N85/582851) where it is poorly welded and again overlies pyroclastics with a basal 70 mm thick pumice ash zone. In this region the base has a transgressive red colouration, variable in thickness, and attributed to post-depositional oxidation during cooling of the ignimbrite flow.

Xenoliths of Rahopaka Ignimbrite in basaltic scoria on Harry Johnson Road (N85/584854) indicate that, like the Tikorangi Ignimbrite, the Rahopaka Ignimbrite extends south at depth underlying the Marshall Ignimbrite and the Huka Group sediments within the Guthrie Graben.

Three ignimbrites and an associated fine pisolitic tuff followed by interbedded slump breccias and stratified lake silts (Fig. 1) are inferred to be younger than the Rahopaka Ignimbrite. This sequence is overlain by the Marshall Ignimbrite.

MARSHALL IGNIMBRITE (ms1–4) (revised name)

This formation was named "Marshall Ignimbrites" by Martin (1961) and is here renamed "Marshall Ignimbrite". Martin designated a type locality in the valley extending north from Marshall's Lookout near Mangakino (N85/210790) and divided the formation into 4 members, 3 of which were given informal lithological names, and the fourth formally designated the Taupaki Ignimbrite or Member (Fig. 2).

Martin recognised the Taupaki Ignimbrite as a key marker bed in the Taupo Volcanic Zone and described its occurrence along the southern and southwestern margins of the Mamaku (Patatere) Plateau where it underlies the Waiotapu and Mamaku Ignimbrites. However, field investigations in the Matahana area indicated that Martin's

"Taupaki Ignimbrite" appears to represent only the welded part of the Marshall Ignimbrite, and is gradational into an upper and lower non-welded zone. As these unwelded "pumice breccias" and the "Taupaki Ignimbrite" are all products of the same eruption, it is felt that they should be classed as 1 ignimbrite.

At the type section at Mangakino, the lower "tuff-breccia ignimbrite" and the upper "pumice tuff breccia" of Martin are correlatives of the upper and lower non- or poorly welded sub-units recognised in the Matahana Basin, that is, the 3 members of Martin's "Marshall Ignimbrites" may be classified as 1 ignimbrite, this being generally non-welded but containing a central welded zone.

The basal lenticulite (which was not observed at the type section and is not present within the Matahana Basin) is now excluded. It is possible that this lenticulite, which Martin considered to be texturally similar to the "Taupaki Ignimbrite" (although more densely welded) is a correlative of the pumiceous Pukerimu Ignimbrite underlying the Marshall Ignimbrite in the Matahana Basin. However, although the Pukerimu Ignimbrite does have a textural resemblance to the welded zone of the Marshall Ignimbrite, it is nowhere seen to become lenticular with depth nor to be totally non-welded at the top. More detailed fieldwork is necessary to resolve the stratigraphy in the Mangakino area.

Detailed descriptions of the Marshall Ignimbrite are given by Martin (1961) and Briggs (1973). Murphy (1977) presented a detailed study of zonal variation and source. One particular point of interest, however, is that the ignimbrite has a high proportion of lithic clasts, predominantly rhyolite, but also including Tikorangi and Rahopaka Ignimbrites. The xenoliths are randomly distributed through the ignimbrite and are not confined to the base.

WAIOTAPU IGNIMBRITE (wp)

This was named by Grindley (1959) and defined by Martin (1961). In the Matahana Basin the ignimbrite unconformably overlies the Marshall Ignimbrite and is itself unconformably overlain by the Mamaku Ignimbrite.

MAMAKU IGNIMBRITE (mk)

This was named by Martin (1961) who defined the type locality as the gorge of the Mangarewa Stream at the Kaharoa–Tauranga Road Bridge (N67/646258). Briggs (1973) gives a generalised profile through the ignimbrite at the type area. It crops out in the southwest of the Matahana Basin where it separates into a number of flows. Here it is a crystal-rich quartzose ignimbrite which, apart from a thin zone of dense welding, is generally poorly welded. Phenocrysts consist of plagioclase, quartz, hypersthene, magnetite, and trace amounts of hornblende.

DATING

Paleomagnetism

All available paleomagnetic dates indicate that the ignimbrites of the Central Volcanic Region are Quaternary. Twelve ignimbrite formations of the Taupo–Rotorua district are all normally magnetised, and are referred to the Brunhes Normal Paleomagnetic Epoch (Cox 1969, 1971), that is, <0.72 Ma (scale of McDougall 1979), while the Waitearikl and Whitianga Ignimbrites, and the Ongatiti Ignimbrite and Rangitoto Ignimbrite Group are all reversely magnetised (Cox 1969, 1971; Kohn 1973). These reversed ignimbrites are referred to the late Matuyama Paleomagnetic Epoch (>0.72 Ma).

In the Matahana area, the oldest previously described formation that can be identified is the Marshall Ignimbrite which is normally magnetised (Cox 1971), that is, <0.72 Ma. The 3 newly defined ignimbrites older than the Marshall Ignimbrite were subjected to paleomagnetic analysis.

Method

At least 3 oriented 2-cm-diameter cores were taken at each site. One site for each of the Pukerimu and Tikorangi Ignimbrites, and 3 sites for the Rahopaka Ignimbrite were studied. The NRM of each sample was measured in a modified Schoenstedt spinner magnetometer.

To test the stability of the NRM, at least 1 sample from each site or ignimbrite was partially demagnetised in alternating magnetic fields of 50, 100, 150, 200, 250, and 300 oersteds (Oe). A demagnetisation diagram was produced for each sample by plotting the end point of the resultant magnetisation vector after every step (Zijderveld 1967). In all but 1 case, the resultant magnetisation vector decreased towards the centre of the co-ordinate system and showed a stable magnetic component remaining after removal of a weak secondary viscous remanent magnetisation (VRM). In most test samples the VRM was removed by an alternating field of 150 Oe. All samples were then cleaned at 200 Oe. (Details are given in Murphy 1977.)

Unit	N	DECL	INCL	K	α_{95}	R
Rahopaka Ignimbrite	9	194.2	+47.4	12.3	16.6	8.347
Tikorangi Ignimbrite	3	175.4	+34.7	15.1	22.5	2.868
Pukerimu Ignimbrite	4	172.7	+31.7	42.4	10.9	3.929
Pukerimu–Tikorangi average	7	173.8	+33.0	29.0	11.0	6.793

Table 1 Average directions of magnetisation of ignimbrite units. Samples cleaned in alternating magnetic field of 200 Oe. N = number of samples, DECL = mean declination, INCL = mean inclination, K = precision parameter, α_{95} = radius of circle of 95% confidence, R = vector sum of N unit vectors parallel to directions of magnetisation.

Results

Results are listed in Table 1. The Pukerimu, Tikorangi, and Rahopaka Ignimbrites all have reversed polarity and are therefore older than 0.72 Ma, but not within the Jaramillo Normal Event from 0.89 to 0.94 Ma.

Geological evidence indicates that the Pukerimu and Tikorangi Ignimbrites form a compound cooling unit and they should thus have the same direction of magnetisation; this is confirmed by the overlap of the cones of confidence (95% level) of the average directions of their magnetisation (Fig. 3).

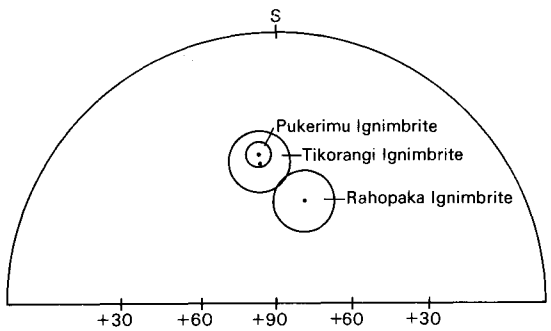


Fig. 3 Directions of magnetisation (cleaned 200 Oe) plotted on the lower hemisphere of an equal area projection. The circles represent 95% cones of confidence.

Fission track dating

For each ignimbrite, approximately 2 kg of sample was processed. The sample was wet sieved and the 64–250 μm fraction retained. Zircon grains finer than 64 μm are too small to handle; zircon grains greater than 250 μm are most unlikely to be found in an ignimbrite. An initial heavy concentration was made using a gold pan, and further separated using heavy liquids and the Franz Isodynamic separator.

The zircons were mounted in teflon sheet, polished and etched for more than 70 h in molten KOH-NaOH solution at 200°C. In the early preparations many zircons fell from the teflon sheet, probably because they were rimmed with glass which was rapidly dissolved by the etchant. Subsequently, zircons were treated in 40% HF at room temperature to remove this glass.

The samples, with mica overlays, were irradiated with NBS standard SRM 962 (Carpenter & Reimer 1974). After irradiation, the spontaneous tracks were counted in the zircon crystals (on prism faces only) and the induced tracks were counted on the mica which was etched for 13 min at room temperature in 48% HF. Essentially, individual crystal ages can be calculated because the same area is counted on the mica overlay for each zircon. However, with such young material, the track density statistics on individual crystals would be meaningless because many crystals (particularly in the Mamaku Ignimbrite) had only 1 spontaneous track, and often none. For young material (<1 Ma), fission track dating is time consuming, but it appears to be the most satisfactory technique which spans the age range between ^{14}C and K/Ar dating techniques.

Although there is controversy regarding the most accurate decay constant for spontaneous fission of ^{238}U, the value used here is the same as previously used in New Zealand studies (i.e., 6.85×10^{-17}/y; Fleischer & Price 1964). When international agreement is reached on the decay constant, then all New Zealand dates can, if necessary, be readily recalculated.

Results of analyses are presented in Table 2. The ages are consistent with the paleomagnetic interpretation. The Tikorangi and Rahopaka Ignimbrites which are paleomagnetically reversed are 1.03 and 0.72 Ma respectively and are within the Matuyama Reversed Epoch. The Marshall and Mamaku Ignimbrite ages are consistent with eruption during the Brunhes Normal Epoch.

DISCUSSION

The fission track ages and the paleomagnetic stratigraphy, summarised in Table 3, indicate that the oldest ignimbrites mapped in the Matahana Basin were deposited about 1 m.y. ago. Although the whole ignimbrite sequence of the central North Island is not fully correlated from area to area, there is growing evidence that the major phase of volcanism commenced a little more than 1 m.y. ago.

Table 2 Fission track data for ages determined on zircon crystals from 4 ignimbrites in the Matahana Basin. Dosimetry standards: NBS SRM962 Cu calibration. $\lambda_F = 6.85 \times 10^{-17}$/y.

Ignimbrite	No. of grains counted	Spontaneous tracks (ρs) counted	Induced tracks (ρi) counted	Neutron dose (ϕ) $\times 10^{15}$ n/cm^2	Age T $\times 10^6$ y	Error $\pm 2\sigma$ $\times 10^6$ y
Mamaku	15	15	3107	0.95 (3000)*	0.14	0.08
Marshall	14	53	2335	0.75 (4046)*	0.52	0.14
Rahopaka	45	78	2279	0.69 (5500)*	0.72	0.16
Tikorangi	39	130	3494	0.90 (7000)*	1.03	0.18

*Number of tracks counted to determine track density.

Table 3 Paleomagnetic stratigraphy and fission track ages of pyroclastics in the Matahana Basin.

	Direction of magnetisation	Fission track age
Mamaku Ignimbrite	Normal*	0.14 ± 0.08
Waiotapu Ignimbrite	Normal*	—
Marshall Ignimbrite	Normal*	0.52 ± 0.14
Rahopaka Ignimbrite	Reversed	0.72 ± 0.16
Tikorangi Ignimbrite	Reversed	1.03 ± 0.18
Pukerimu Ignimbrite	Reversed	1.03 ± 0.18

*from Cox (1969)

This is further confirmed by ages of distal tephras (Ninkovich 1968; Seward 1974, 1975, 1976; Watkins & Huang 1977) in marine sequences.

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